

Deadline: Jan 26

Solutions will be uploaded after the deadline

1. Suppose $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$.

(a) Show that $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}$.

(b) Show that $\sum_{i=1}^n (x_i - \bar{x})$ is *always* equal to 0.

2. Consider a simple linear regression model without an intercept term, i.e., $y_i = \beta_1 x_i + u_i$. Let $\{(x_i, y_i): i = 1, \dots, n\}$ denote a random sample of size n from the population. Derive the OLS slope estimator $\hat{\beta}_1$.

3. Consider the estimated OLS regression model for consumption function, $\ln(\widehat{\text{consumption}}) = 3 + 0.15 \text{disposable_income} - 0.065 \text{interest_rate}$, where interest rate is measured in % and disposable income is measured in thousands of dollars. Interpret the estimate $\hat{\beta}_{\text{disposable_income}} = 0.15$ and $\hat{\beta}_{\text{interest_rate}} = 0.065$.

4. Consider a simple regression model. Suppose $\hat{\beta}_1 = 2.5$ and $\text{se}(\hat{\beta}_1) = 0.3$ are based on a sample of size 200 from the population.

(a) Construct a 95% confidence interval for β_1 . Explain this confidence interval in details.

(b) Construct a 99% confidence interval for β_1 .